

DURATION: 2 hours 30 mins

MAXIMUM MARKS: 75

- Note:
1. All questions are compulsory with internal choice.
 2. Draw neat diagrams wherever necessary.
 3. Figures to the right indicate full marks.

QI Attempt any three of the following:

[15]

1. Describe the 3D's of security and their significance.
2. Explain the CIA triad and its role in information security.
3. Justify the following statement with respect to information security, "The Job of a defender is harder than the job of an attacker".
4. Justify the investment in security with business benefits.
5. Compare and contrast the Lollipop and Onion security models with diagrams.
6. Describe two types of security methodologies.

QII Attempt any three of the following:

[15]

1. Describe how Kerberos authentication works with a diagram.
2. Enlist and explain different types of authorization systems.
3. Demonstrate RSA encryption with an example.
4. Define cipher. Compare transposition and substitution ciphers with an example.
5. Discuss the best practices for storage security.
6. Analyze database object-level security measures.

QIII Attempt any three of the following:

[15]

1. Describe the following concepts with respect to network availability
 - a. Hot standby
 - b. Cold standby
2. Explain Network Address Translation (NAT) with an example.
3. Discuss the features of firewalls.
4. Describe the roles of hubs and switches in network security.
5. Identify two types of wireless attacks and their impacts.
6. Evaluate the best practices for network hardening.

QIV Attempt *any three* of the following:

[15]

1. Define IDS and IPS. Explain the types of IDS alerts with an example.
2. Evaluate the impact of Security Information and Event Management system.
3. Compare host-based and network-based IDS.
4. Describe the concept of PBX and the importance of PBX security.
5. Explain the concept of Mandatory Access Control (MAC) and Discretionary Access Control (DAC).
6. Discuss the concept of reference monitor and Reference Computing model.

QV Attempt *any three* of the following:

[15]

1. Define a hypervisor and explain its security concerns.
2. Explain cloud computing and types of cloud services.
3. Differentiate between public and private cloud.
4. Discuss remote administration security, its pros and cons.
5. Categorized various corporate physical assets for better security.
6. Describe the various factors for securing the assets with physical security devices.

Note: 1. All questions are compulsory with internal choice.

2. Draw neat diagrams wherever necessary.

3. Figures to the right indicate full marks.

QI Attempt any three of the following: [15]

- Define BI. Describe Effective and Timely decisions.
- Write a note on Ethics and Business Intelligence.
- Illustrate components of Business Intelligence architecture (Basic building blocks) with a neat diagram.
- Define System. Compare open and closed system. Describe two main classes of evaluation metrics of system.
- Describe the aspects characterizing a decision-making process.
- Classify the types of Decisions.

QII Attempt any three of the following: [15]

- Outline the phases in the development of mathematical models for decision making with a neat diagram.
- Categorize the types of models based on characteristics and probabilistic nature.
- Describe the data mining process.
- Explain Dataset and categories of dataset.
- Write a note on Data Transformation and Reduction.
- Enlist and explain the techniques in Data Validation.

QIII Attempt any three of the following: [15]

- Compare the Euclidean distance, Manhattan distance and Minkowski distance.
- Explain the K-means algorithm and describe its steps.
- Write a note on Hierarchical methods.
- Outline the steps in k-medoids algorithm.
- Consider the following set of 6 one dimensional data points: 18,22,25,42,27,43
Apply the agglomerative hierarchical clustering algorithm to build the dendrogram. Clearly show the proximity matrix corresponding to each iteration of the algorithm.
- Interpret the Divisive Hierarchical methods and evaluation of clustering models.

QIV Attempt any three of the following:

[15]

- a. Enlist and explain the key characteristics of high-quality information.
- b. Contrast the cost of sales actions and their revenue in relational marketing. Also explain customization in relational marketing.
- c. Explain the Lifetime Value of a customer with a neat diagram.
- d. Elaborate on Salesforce Management.
- e. Relate and explain different Optimization models for logistics planning.
- f. Describe Revenue management and decision processes in revenue management.

QV Attempt any three of the following:

[15]

- a. Define Knowledge. Explain Knowledge Management System and KM Metrics.
- b. Explain Organizational maturity model.
- c. Enlist and explain the characteristics of Artificial Intelligence.
- d. Compare and contrast AI versus Natural Intelligence.
- e. What is Machine Learning? How do machines learn? Explain with a neat diagram.
- f. Elaborate Machine Learning tools.